Intelligent Association

Presented by

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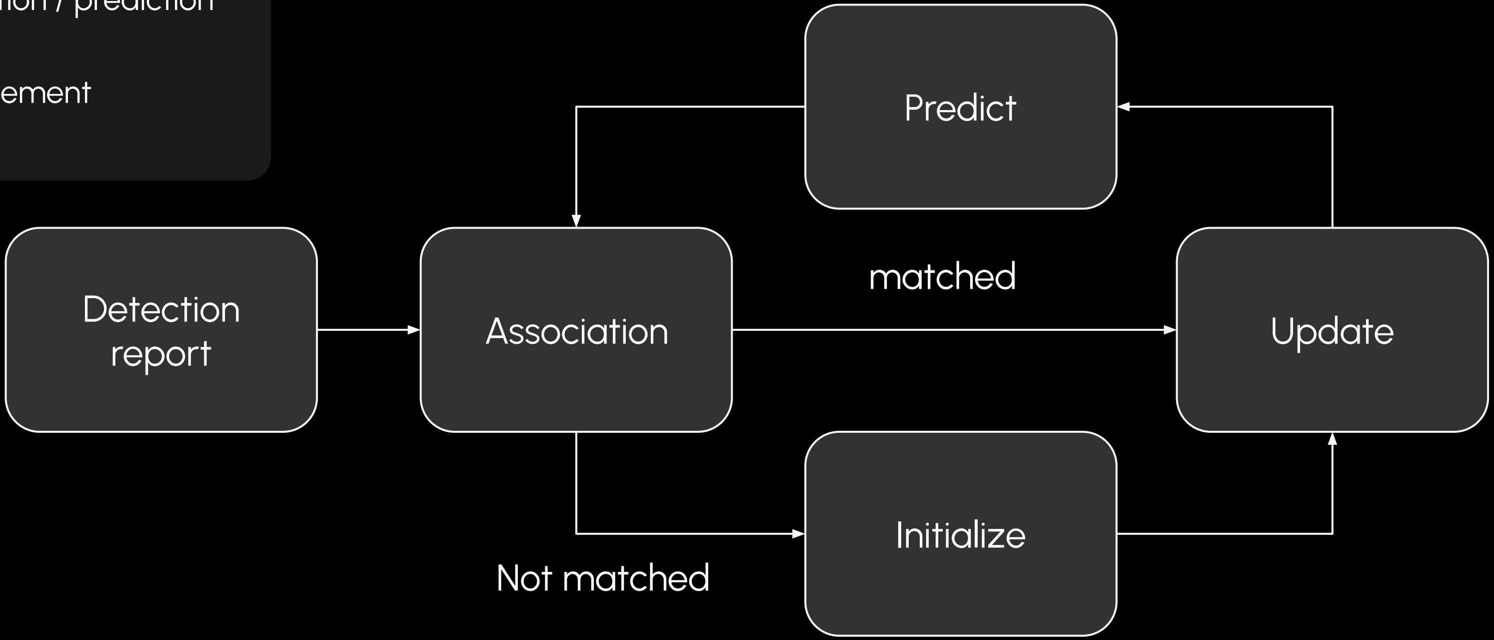
Background - Tracking

Three main parts of tracking:

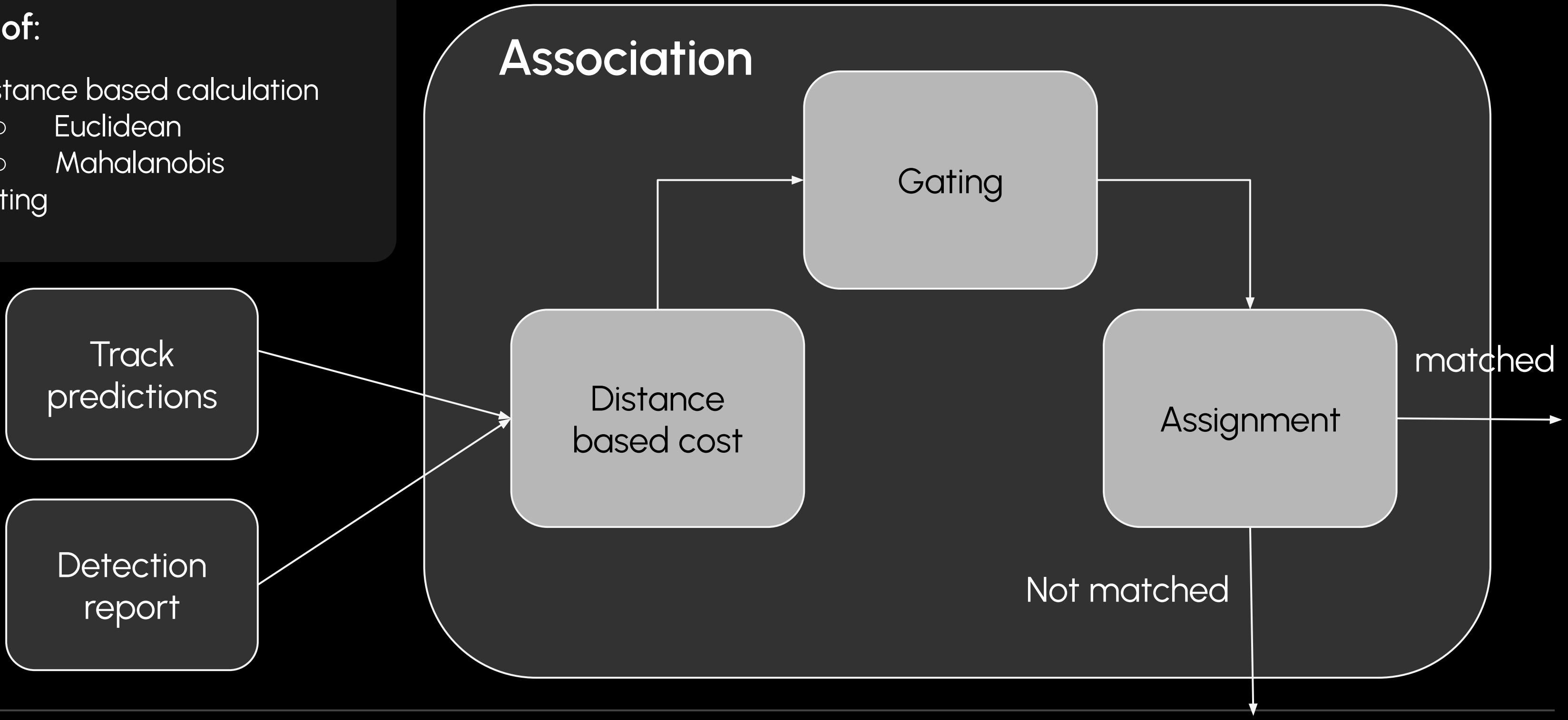
- State estimation / prediction
- Association
- Track Management



Background - Classical Association

Built off of:

- Distance based calculation
  - Euclidean
  - Mahalanobis
- gating



Motivation - Noise

Uncertainty or error introduced into the system

System level:
- Measurement noise

Detection level:
- False positives
- False negatives

Classical Association
- Not robust
  - Optimize Gating?
- Wastes data
- Simple relationships

Direct Impact
- Swapped tracks

Error propagation
- Worse predictions
- Decreased stability

Problem:

Classical distance based association is not robust to noise

Solution:

Can intelligent ML model improve association

Solution :

Simulate:

FMCW radar detections

Train:

Model to score track, detection pair

Replace:

Place model in tracking system

Provide feature specific domain:

- Power
- Radial velocity
- ect

Exploration:

- Feature space
- MLP -> layers, neurons ...

Changes to existing architecture:

- Distance -> probability
- gating-> thresholding

Evaluation

Model Level:
- F1
- generalizability

Tracking Level:
Does intelligent association improve tracking systems

- MOTA
- MOTP
- Average Swaps
- Average frags
- FPS

Questions?